Python Programming with Data Science – Comprehensive Syllabus

Course Overview

This course introduces Python programming from the basics to advanced data science concepts. Students will learn programming fundamentals, data manipulation, visualization, statistical analysis, machine learning, and project development using industry-standard Python libraries.

Duration: 12–16 Weeks (60–80 Hours)

Prerequisites:

- Basic computer knowledge
- High school mathematics (recommended)
- No prior programming experience required

Module 1: Introduction to Python (Week 1)

Topics

- Introduction to Python
- Installing Python and Anaconda
- Using Jupyter Notebook and VS Code
- Python syntax
- Variables and data types
- Input and output
- Comments and documentation

Practical Exercises

- Install Python environment
- Write simple Python programs
- Temperature converter
- Basic calculator

Module 2: Operators and Control Statements (Week 2)

Topics

- Arithmetic operators
- Comparison operators
- Logical operators
- Assignment operators
- Membership and identity operators
- if, elif, else
- Nested conditions

Practical Exercises

- Grade calculator
- Leap year checker
- BMI calculator

Module 3: Loops and Functions (Week 3)

Topics

- for loop
- while loop
- break, continue, pass
- Functions
- Parameters
- Return values
- Lambda functions
- Scope of variables

Practical Exercises

- Prime number checker
- Fibonacci series
- Factorial program
- Number guessing game

Module 4: Data Structures (Week 4)

Topics

- Lists
- Tuples
- Dictionaries
- Sets
- String operations
- List comprehension

Practical Exercises

- Student record management
- Word frequency counter
- Dictionary-based phonebook

Module 5: File Handling & Exception Handling (Week 5)

Topics

- Reading files
- Writing files
- CSV files
- JSON files
- Exception handling
- Try, Except, Finally
- Custom exceptions

Practical Exercises

- Student marks file
- CSV reader
- Error handling examples

Module 6: Object-Oriented Programming (Week 6)

Topics

- Classes
- Objects
- Constructors
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction

Practical Exercises

- Employee management system
- Bank account simulation
- Library management system

Module 7: Python Libraries (Week 7)

Topics

- NumPy
- Pandas
- Matplotlib
- Seaborn

Practical Exercises

- Array operations
- DataFrames
- Statistical summaries
- Basic visualizations

Module 8: Data Analysis with Pandas (Week 8)

Topics

- Importing datasets
- Cleaning data
- Handling missing values
- Data filtering
- Grouping
- Aggregation
- Merging datasets

Practical Exercises

- Analyze sales dataset
- Employee dataset
- Student performance analysis

Module 9: Data Visualization (Week 9)

Topics

- Line charts
- Bar charts
- Pie charts
- Histograms
- Scatter plots
- Box plots
- Heatmaps

Practical Exercises

- Sales dashboard
- COVID-19 visualization
- Population analysis

Module 10: Statistics for Data Science (Week 10)

Topics

- Mean

- Median
- Mode
- Variance
- Standard deviation
- Correlation
- Probability basics
- Normal distribution

Practical Exercises

- Statistical analysis using Python
- Correlation study
- Distribution visualization

Module 11: Introduction to Machine Learning (Week 11)

Topics

- Machine learning overview
- Supervised learning
- Unsupervised learning
- Model training
- Train-test split
- Evaluation metrics

Libraries

- Scikit-learn

Practical Exercises

- House price prediction
- Student score prediction
- Iris dataset classification

Module 12: Machine Learning Algorithms (Week 12)

Topics

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- K-Nearest Neighbors (KNN)
- K-Means Clustering
- Naive Bayes

Practical Exercises

- Customer segmentation
- Email spam detection
- Diabetes prediction

Module 13: Mini Project (Week 13)

Choose one:

- Sales Analysis Dashboard
- Employee Analytics
- Student Performance Analysis
- Movie Recommendation
- Weather Data Analysis
- Stock Market Analysis
- COVID-19 Data Dashboard

Module 14: Capstone Project (Week 14–16)

Complete Project Lifecycle

- Problem identification
- Data collection
- Data cleaning
- Exploratory Data Analysis (EDA)
- Feature engineering
- Model building
- Model evaluation

- Visualization
- Final report
- Presentation

Tools & Libraries

- Python
- Jupyter Notebook
- VS Code
- Anaconda
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- SciPy (optional)
- Plotly (optional)

Assessment Scheme

Component	Weightage
Assignments	20%
Lab Exercises	20%
Quizzes	10%
Mini Project	20%
Capstone Project	30%

Learning Outcomes

By the end of the course, learners will be able to:

- Write Python programs using core programming concepts.
- Use Python data structures and object-oriented programming techniques.

- Read, clean, transform, and analyze datasets with Pandas.
- Create informative visualizations using Matplotlib and Seaborn.
- Apply descriptive statistics to interpret data.
- Build and evaluate basic machine learning models with Scikit-learn.
- Complete end-to-end data science projects following industry practices.
- Present analytical findings through reports and visual dashboards.